Investment Irreversibility in a Granular World*

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Abstract

We show that investment irreversibility plays a prominent role in shaping aggregate fluctuations in a granular economy, characterized by a fat-tailed distribution of non-atomistic firms. Such an economy satisfies two conditions necessary for irreversibility to matter for aggregate fluctuations: idiosyncratic shocks volatile enough to bind the irreversibility constraint cyclically for a mass of large firms, and a failure of the law of large numbers that prevents firm-level fluctuations from washing out in the aggregate. Through both a direct misallocation channel and an indirect general equilibrium channel, irreversibility dampens the volatility of output, investment, and hours while amplifying the volatility of consumption. Eliminating the friction yields a consumption-equivalent welfare gain of 1.83%; however, merely subsidizing capital liquidation reduces welfare by 0.59%, because firms, compensated by the government, do not internalize the aggregate resource losses and therefore over-disinvest.

Keywords. Investment irreversibility; Business cycles; Idiosyncratic shocks; Misallocation

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1 Introduction

Investment irreversibility is a well-documented source of capital misallocation.[1] This friction can be particularly consequential in recessions, when many firms wish to downsize but face steep discounts in liquidating equipment and machinery, freezing capital reallocation across firms.[2] It has long motivated policy intervention, most prominently the 2009 restructurings of GM and Chrysler to facilitate capital reallocation.[3] While the effects of this friction on firm-level investment are well documented, whether it also matters at the aggregate level—and therefore whether policy interventions targeted at it can be welfare-enhancing—depends on its role in aggregate fluctuations. This paper asks when investment irreversibility plays a prominent role in aggregate fluctuations, how large that role is, and what it implies for policies targeted at this friction.

We answer that two conditions must hold: the irreversibility constraint must bind cyclically for a mass of large firms, and the resulting firm-level fluctuations must not wash out in aggregation. Both conditions are satisfied in a granular economy, where the firm size distribution is fat-tailed and firms are not atomistic, so the law of large numbers fails and idiosyncratic shocks to large firms do not average out in the aggregate as in Gabaix (2011), Di Giovanni, Levchenko and Méjean (2014), and Carvalho and Grassi (2019).[4] To evaluate this quantitatively, we build a heterogeneous-firm general equilibrium business cycle model

[1] Seminal works include Hsieh and Klenow (2010), Restuccia and Rogerson (2008), and David and Venkateswaran (2019).

[2] Investment irreversibility arises from multiple sources including asset specificity (Ramey and Shapiro, 2001; Lanteri, 2018), technological obsolescence (Caunedo and Keller, 2021), and thin resale markets (Ottonello, 2026). Kermani and Ma (2023) document that even general-purpose equipment and machinery typically recover only 50–70% of replacement cost upon disposal, while specialized assets face even higher losses of 10–30%.

[3] Section 363 sales in bankruptcy law allow expedited asset transfers to preserve going-concern value, enabling rapid transfer of viable assets to "New GM" and the Fiat consortium while avoiding piecemeal liquidation (Brubaker and Tabb, 2010; Lubben, 2009; Roe and Skeel, 2010).

[4] Gabaix (2011) shows that when the firm size distribution is sufficiently fat-tailed, idiosyncratic shocks to a small number of large firms fail to average out by the law of large numbers, and they generate aggregate fluctuations on their own — granular fluctuations. Carvalho and Grassi (2019) embed this mechanism in a heterogeneous-firm RBC model, showing that granular shocks generate realistic business-cycle moments. Di Giovanni et al. (2014) provide direct micro-level evidence, finding that firm-specific shocks account for a substantial share of aggregate volatility.

with investment irreversibility, calibrated to match the U.S. firm-size distribution—most notably its highly skewed right tail—and driven solely by idiosyncratic productivity shocks. We find that investment irreversibility substantially dampens the volatility of aggregate output, investment, and hours worked, while amplifying consumption volatility. Building on these findings, we show that fully removing the friction would yield substantial welfare gains in consumption-equivalent terms, but liquidation subsidies—a widely adopted policy response—reduce welfare, as firms fail to internalize the aggregate resource cost of excessive disinvestment.[5]

Why these two conditions? In models with atomistic firms, idiosyncratic shocks wash out by the law of large numbers, so the constraint cannot cycle from idiosyncratic variation. Aggregate TFP shocks are themselves insufficiently volatile to cycle the constraint either (Veracierto, 2002). Granularity breaks this bind: the idiosyncratic shocks that hit large firms in a fat-tailed size distribution are an order of magnitude more volatile than aggregate TFP and do not average out. We verify this in the paper — subjecting the same granular firms to aggregate TFP shocks alone, the fraction of cyclically constrained firms collapses and the propagation mechanism disappears. We trace this dampening of aggregate volatility to two channels: a direct misallocation channel and an indirect, general equilibrium channel.

A direct channel stems from the impact of irreversibility on capital allocation. Central to our analysis is that aggregate output volatility is positively related to the cross-sectional covariance between capital and productivity: when capital is concentrated in high-productivity firms, aggregate output fluctuates more.[6] Irreversibility causes firms' capital stocks to adjust more sluggishly to shocks, substantially dampening this covariance and reducing the volatility of aggregate output, investment, and hours worked.

An indirect general equilibrium channel operates through factor prices. When capital

[5]Examples of such policies include Japan's shipbuilding capacity reduction under the 1978 Structurally Depressed Industries law, which paid for permanent scrapping of shipyards (United States General Accounting Office, 1983), and the UK's coal mine closure programs that provided redundancy payments to accelerate pit shutdowns (Glyn, 1992).

[6]Appendix A formalizes this: in a partial-equilibrium model with finitely many firms, the coefficient of variation of aggregate output is increasing in the productivity–capital covariance

adjustment is restricted by irreversibility, the margin firms would normally use to cushion shocks — adjusting their capital stock — is constrained, and adjustment is forced onto labor and asset markets in general equilibrium. Real wages become more volatile, falling more sharply during downturns; this acts as a shock absorber for firms, mitigating the fall in employment and production and further dampening aggregate output and hours volatility.

The same constraint propagates into the consumption-savings margin. Aggregate consumption is more volatile and the real interest rate is countercyclical — patterns the standard frictionless real business cycle model fails to deliver, and that investment irreversibility generates here.[7] These price patterns reflect two underlying costs: less smoothed consumption and resource losses from costly disinvestment.

These welfare costs are substantial and unambiguous. Eliminating the friction altogether yields a consumption-equivalent gain of 1.83%, driven by gains on both the production and household sides: capital is allocated more productively across firms and is no longer destroyed in costly disinvestment, while households smooth consumption more effectively. We compare this frictionless benchmark to a conventional policy response — subsidizing liquidation losses — which reduces welfare by 0.59%. The subsidy lowers the private cost of disinvestment without lowering its social cost, so firms disinvest more when subsidized than when not, and more than the available welfare gain leaks out as resource waste.[8] These results carry a cautionary message: real-world interventions — from the Troubled Asset Relief Program in the U.S. to capacity reduction programs implemented across many countries — should target the friction itself rather than compensate firms for bearing its costs.

Taken together, our results characterize the aggregate dynamics — volatilities and comovements — of a granular economy with investment irreversibility, along with the welfare

[7]The asset-pricing literature has long established that generating these patterns requires departing from the frictionless benchmark, either on the household side through habit formation (Campbell and Cochrane, 1999) or on the production side through capital adjustment costs (Jermann, 1998; Beaudry and Guay, 1996).

[8]Two strands of the literature speak to this failure. Bianchi and Mendoza (2018) is about the wedge: private agents do not internalize a social cost, and a policy that closes the wedge for agents leaves it open at the aggregate level. Farhi and Tirole (2012) is about behavioral feedback: the policy itself induces more of the costly behavior. Our subsidy combines both.

implications of policies that attempt to alleviate the irreversibility constraint faced by firms. The analysis is made possible by a solution method we develop for granular general equilibrium models under aggregate uncertainty, building on the multivariate discretization of Terry and Knotek II (2011). The method extends naturally to the study of other real or financial frictions in granular economies.

Related literature

This paper contributes to the literature on how firms' idiosyncratic shocks generate aggregate fluctuations (Jovanovic, 1987; Gabaix, 2011; Gabaix and Koijen, 2024; Carvalho and Gabaix, 2013; Carvalho and Grassi, 2019; Di Giovanni and Levchenko, 2012; Di Giovanni et al., 2014; Di Giovanni, Levchenko and Méjean, 2018, 2024; Yeh, 2025; Ifergane, 2024; Burstein, Carvalho and Grassi, 2025). A distinct feature of our study is the inclusion of real frictions and capital misallocation—elements typically absent from granular models.[9] Our framework is unique in extending a general equilibrium business cycle model to incorporate both granular firms and investment irreversibility, and our solution method is applicable more broadly to the study of any real or financial friction in granular economies.

This paper also contributes to the literature on the macroeconomic implications of micro-level investment frictions, particularly investment irreversibility.[10] A range of mechanisms have been shown to make the friction matter at the aggregate level, including endogenous used-capital prices (Lanteri, 2018) and financial frictions (Khan and Thomas, 2013; Cui, 2022).[11] These all operate within models in which firms are atomistic. Our contribution is

[9] In Appendix H, we also provide empirical evidence that granular shocks generate dispersion in the marginal product of capital.

[10] Seminal partial-equilibrium analyses at the firm level include Pindyck (1988), Abel and Eberly (1994), Dixit and Pindyck (1994), and Bertola (1998). Cross-sectional aggregation results include Bertola and Caballero (1994) on dampening of aggregate investment, Abel and Eberly (1996) on asymmetric persistence, Caballero and Engel (1999) on the generalized (S,s) framework, and Baley and Blanco (2026) on amplification through the cross-sectional invariant distribution. Cooper and Haltiwanger (2006) provide the leading plant-level structural estimation.

[11] Other routes include Bloom (2009) and Bloom, Floetotto, Jaimovich, Saporta-Eksten and Terry (2018) on time-varying dispersion of idiosyncratic shocks; Fiori (2012) on multi-sector capital heterogeneity and Fiori and Scoccianti (2026) on embodied technology vintages; Gourio and Kashyap (2007) and Lee (2022)

distinct: we identify granularity itself — a fat-tailed firm size distribution in which idiosyncratic shocks to large firms drive aggregate fluctuations — as a route by which firm-level irreversibility alters the volatilities and cross-correlations of aggregate output, investment, hours, and consumption.

Finally, this paper is connected to the misallocation literature (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2010; David and Venkateswaran, 2019) and we especially advance research on capital misallocation in business cycles (Bloom et al., 2018; Khan and Thomas, 2013; Lanteri, 2018). We find that the covariance between firm-level productivity and capital is positively related to aggregate output volatility—a key mechanism for understanding the stability of the aggregate economy and the role of frictions such as investment irreversibility.

Organization

The rest of the paper is organized as follows. Section 2 develops the general equilibrium model and its solution method. Section 3 presents the model's calibration and quantitative results. Section 4 discusses policy implications and we conclude in Section 5.

2 General Equilibrium Model

This section develops a quantitative heterogeneous firm business cycle model to assess the impact of idiosyncratic shocks and investment irreversibility on capital misallocation and aggregate fluctuations. We build upon the standard framework where firms use capital and labor to produce a homogeneous good, and where capital is chosen one period in advance. The key departure from the standard model is that we allow the distribution of firm-level productivities to fluctuate stochastically over time, even though the underlying ergodic productivity process for individual firms is a standard Markov chain. This allows us to study

on the cross-sectional state of large plants near the inaction boundary; Bachmann, Caballero and Engel (2013) on parameter dependence; and House (2014), Winberry (2021), and Koby and Wolf (2020) on the price-elasticity conditions under which cross-sectional heterogeneity affects aggregate dynamics.

the effects of idiosyncratic shocks that do not average out in the aggregate. Importantly, we maintain a rational expectations framework: agents understand the stochastic process governing the evolution of the productivity distribution and form expectations accordingly. This allows us to use recursive methods for the analysis.

2.1 Production and Investment

There is a finite number, M, of firms in the model economy. Each firm undertakes production using a predetermined capital stock k, and labor n. Firms are heterogeneous in their idiosyncratic productivity, ε. The ergodic process for idiosyncratic productivity, ε, is a Markov chain with a finite number of states: $\varepsilon \in \{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_{N_\varepsilon - 1}, \varepsilon_{N_\varepsilon}\}$. The transition probabilities for this individual firm productivity process are given by $Pr(\varepsilon' = \varepsilon_j | \varepsilon = \varepsilon_i) \equiv \Pi^\varepsilon(\varepsilon_i, \varepsilon_j) \geq 0$, with the standard property that $\sum_{j=1}^{N_\varepsilon} \Pi^\varepsilon(\varepsilon_i, \varepsilon_j) = 1$ for all $i = 1, 2, \ldots, N_\varepsilon$.

At the beginning of each period, firms observe the current aggregate state of the economy and choose their current level of employment, n. Production then takes place. After production, firms make their investment decisions, subject to a partial irreversibility friction, represented by the parameter ψ.

The aggregate state of the economy is characterized by the distribution of firms across idiosyncratic productivity and capital, denoted by $\mu(\varepsilon, k)$. This distribution is defined over the space $\mathbf{S} = \mathbf{E} \times \mathbb{R}_+$, where $\mathbf{E} = \{\varepsilon_1, \ldots, \varepsilon_{N_\varepsilon}\}$. Unlike standard models where this distribution is stationary, we allow $\mu(\varepsilon, k)$ to vary stochastically over time. This captures the idea that idiosyncratic shocks do not perfectly cancel out in the cross-section due to the finite number of firms. The evolution of μ is the sole source of aggregate uncertainty in the model. We model this evolution as a Markov process, with $\mu' \sim G(\mu)$, meaning that the distribution in the next period, μ', is drawn from a distribution G that depends on the current distribution, μ.

Crucially, we construct a conditional transition probability for the idiosyncratic productivity process, conditional on both the current and future aggregate state: $Pr(\varepsilon' = \varepsilon_j | \varepsilon =$

$\varepsilon_i, \mu, \mu') \equiv \Pi^{\varepsilon}_{\mu'|\mu}(\varepsilon_i, \varepsilon_j) \geq 0$. This transition probability also satisfies $\sum_{j=1}^{N_\varepsilon} \Pi^{\varepsilon}_{\mu'|\mu}(\varepsilon_i, \varepsilon_j) = 1$ for all $i = 1, 2, \ldots, N_\varepsilon$. The transition probability $\Pi^{\varepsilon}_{\mu'|\mu}(\varepsilon_i, \varepsilon_j)$ depends on both μ and μ' because, while the underlying ergodic process for ε is exogenous, the realized distribution of firms across productivity levels in the next period depends on the evolution of the aggregate state.[12] With a finite number of firms, the realized fraction of firms transitioning from ε_i to ε_j differs from $\Pi^{\varepsilon}(\varepsilon_i, \varepsilon_j)$ — the law of large numbers does not apply — and, conditional on the realized pair (μ, μ'), the transition probability of an individual firm is this realized fraction rather than the ergodic probability. Averaging across realizations of the future aggregate state recovers the primitive chain, $\sum_{\mu'} \Pi^{\mu}(\mu'|\mu)\, \Pi^{\varepsilon}_{\mu'|\mu}(\varepsilon_i, \varepsilon_j) = \Pi^{\varepsilon}(\varepsilon_i, \varepsilon_j)$, where $\Pi^{\mu}(\mu'|\mu)$ denotes the transition probability of the aggregate state implied by G.

We can now state the firm's dynamic optimization problem. Let $V(\varepsilon, k; \mu)$ be the value of a firm at the beginning of a period with idiosyncratic productivity ε and predetermined capital stock k, when the aggregate state of the economy is μ. Firms discount future values using a state-contingent discount factor $d(\mu, \mu')$. The firm's problem is:

$$V(\varepsilon, k; \mu) = \max_{n, k'} \varepsilon F(k, n) - \omega(\mu) n - i + \sum_{\mu'} \Pi^{\mu}(\mu'|\mu) d(\mu, \mu') \sum_{\varepsilon'} \Pi^{\varepsilon}_{\mu'|\mu}(\varepsilon'|\varepsilon) V(\varepsilon', k'; \mu') \quad (1)$$

subject to

$$\hat{i} = k' - (1 - \delta)k \quad (2)$$

$$i = \hat{i}(1 - \psi \mathbf{1}_{\hat{i}<0}) \quad (3)$$

$$k', n \in \mathbb{R}_+ \quad (4)$$

Equation (1) is the Bellman equation. The firm chooses labor, n, and next period's

[12] Imagine a simple example: if μ' represents a state with a much higher concentration of firms at high productivity levels than μ, then the probability of a firm transitioning from a low ε to a high ε' will be higher under $\Pi^{\varepsilon}_{\mu'|\mu}$ than under the ergodic transition matrix Π^{ε}. The probability $\Pi^{\varepsilon}_{\mu'|\mu}$ captures how the aggregate shock (the change from μ to μ') affects the realized transitions of firms between productivity levels. This is distinct from the time-invariant, ergodic transition probabilities Π^{ε}, which govern the underlying stochastic process for an individual firm's productivity.

capital stock, k', to maximize the sum of current profits ($\varepsilon F(k,n) - \omega(\mu)n - i$) and the discounted expected future value. $\omega(\mu)$ is the wage rate, which depends on the aggregate state μ. Equation (2) defines gross investment, $\hat{i}$, as the difference between next period's capital and depreciated current capital. Equation (3) defines investment, i, incorporating the irreversibility friction: if gross investment is negative (disinvestment), the firm receives only a fraction $(1-\psi)$ of the value of the capital it sells. Equation (4) imposes non-negativity constraints on capital and labor. The term $\Pi^{\mu}(\mu'|\mu)$ represents the probability of transitioning from the current aggregate state μ to the future aggregate state μ'.

As in Khan and Thomas (2013), the optimal capital decision rule takes an (S, s) form:

$$K(\varepsilon,k;\mu) = \begin{cases} k_u^\star(\varepsilon,k;\mu), & \text{if } k \leq \frac{k_u^\star(\varepsilon,k;\mu)}{(1-\delta)} \\ k(1-\delta), & \text{if } k \in \left[\frac{k_u^\star(\varepsilon,k;\mu)}{(1-\delta)}, \frac{k_d^\star(\varepsilon,k;\mu)}{(1-\delta)}\right] \\ k_d^\star(\varepsilon,k;\mu), & k \geq \frac{k_d^\star(\varepsilon,k;\mu)}{(1-\delta)} \end{cases} \tag{5}$$

Here, $K(\varepsilon,k;\mu)$ represents the optimal choice of next period's capital stock, given current productivity ε, current capital k, and the aggregate state μ. The firm has two target capital levels: an upward target, $k_u^\star$, and a downward target, $k_d^\star$. If the firm's current capital is below the inaction region (determined by the targets and the depreciation rate), it invests to reach the upward target. If current capital is above the inaction region, it disinvests (subject to the irreversibility constraint) to reach the downward target. If current capital is within the inaction region, the firm simply lets its capital depreciate.

The target capital levels are defined as:

$$k_u^\star = \arg\max_{k'} -k' + \sum_{\mu'} \Pi^{\mu}(\mu'|\mu) d(\mu,\mu') \sum_{\varepsilon'} \Pi^{\varepsilon}_{\mu'|\mu}(\varepsilon'|\varepsilon) V(\varepsilon',k';\mu') \tag{6}$$

$$k_d^\star = \arg\max_{k'} -(1-\psi)k' + \sum_{\mu'} \Pi^{\mu}(\mu'|\mu) d(\mu,\mu') \sum_{\varepsilon'} \Pi^{\varepsilon}_{\mu'|\mu}(\varepsilon'|\varepsilon) V(\varepsilon',k';\mu') \tag{7}$$

Equations (6) and (7) show that the target capital levels are chosen to maximize the expected discounted future value of the firm, taking into account the cost of investment (or

disinvestment) and the transition probabilities for both the aggregate state and the firm's idiosyncratic productivity.

2.2 Households

Our model economy is populated by a unit measure of identical households. Their lifetime expected utility maximization problem is:

$$W(\lambda;\mu) = \max_{c,n^h,\lambda'} U\left(c, 1-n^h\right) + \beta\mathbb{E}\left[W\left(\lambda';\mu'\right)|\mu\right] \tag{8}$$

$$\begin{aligned} \text{subject to:} \qquad & c + \int_{\mathbf{S}} \rho_1\left(\varepsilon', k';\mu\right)\lambda'\left(d\left[\varepsilon\times k'\right]\right) \\ & \leq w(\mu)\, n^h + \int_{\mathbf{S}} \rho_0\left(\varepsilon, k;\mu\right)\lambda\left(d\left[\varepsilon\times k\right]\right). \end{aligned} \tag{9}$$

Households hold one-period shares in firms, denoted by λ. Given the prices—the real wage, $w(\mu)$, and the prices of shares, $\rho_0(\varepsilon, k;\mu)$ and $\rho_1(\varepsilon', k';\mu)$—households choose their current consumption, c, hours worked, n^h, and the number of new shares, λ'. Let $C(\lambda;\mu)$ and $N(\lambda;\mu)$ represent the household decision rules for consumption and hours worked, respectively, and let $\Lambda(\varepsilon, k', \lambda;\mu)$ be the household decision rule for shares purchased in firms that will begin the next period with k' capital and current productivity ε.

2.3 Recursive Equilibrium

A recursive competitive equilibrium is a set of functions:

$$\begin{aligned} \text{prices} &: (\omega, d, \rho_0, \rho_1) \\ \text{quantities} &: (N, K, C, N^h, \Lambda^h) \\ \text{values} &: (V, W) \end{aligned}$$

that solve firm and household problems and clear the markets for assets, labor, and output as follows:

1. V satisfies equation (1), and (N, K) are the associated policy functions for firms.

2. W satisfies equations (8) and (9), and (C, N^h, Λ^h) are the associated policy functions for households.

3. $\Lambda^h(\varepsilon, k; \mu) = \mu(\varepsilon, k)$ for each $(\varepsilon, k) \in \mathcal{S}$. (Asset Market Clearing: Households' shareholdings equal the distribution of firms.)

4. The labor and goods markets clear:

$$\begin{aligned} N^h(\mu) &= \int_{\mathbf{S}} N(\varepsilon, k; \mu) \cdot \mu(d[\varepsilon \times k]) \\ C(\mu) &= \int_{\mathbf{S}} [\varepsilon F(k, N(\varepsilon, k; \mu)) - Q[K(\varepsilon, k; \mu), k] \cdot (K(\varepsilon, k; \mu) - (1-\delta)k)] \cdot \mu(d[\varepsilon \times k]) \end{aligned}$$

   where $Q[K(\varepsilon, k; \mu), k]$ is an indicator function that takes value 1 if $K(\varepsilon, k; \mu) > (1-\delta)k$, and value $(1-\psi)$ otherwise.

5. The resulting individual decision rules for firms and households are consistent with the conditional distribution of the aggregate state, Π^{μ}.

2.4 Solution Method

The computational algorithm builds upon the solution method of Khan and Thomas (2008), itself based on Krusell and Smith (1998). It involves approximating the aggregate state space and solving the firm's problem recursively. A key element is the construction of a Markov chain to approximate the stochastic process for the distribution of firms across productivity transitions.

The main challenge to solving traditional heterogeneous agent model is that the aggregate state vector $\mu(\varepsilon, k)$ contains the cross-sectional distribution of firms, which is an infinite-dimensional object. Krusell and Smith (1998) and Khan and Thomas (2008) approximate the intractable cross-sectional distribution μ with a finite dimensional aggregate state variable, such as some moments of the distribution of capital, to make the model solution computable.

However, the solution of granular economy models must confront an additional key challenge: the cross-sectional distribution μ is not only an infinite-dimensional entity but also inherently stochastic. The failure of the law of large numbers induces stochastic fluctuations in the marginal distribution of firms over productivity, $\mu_\varepsilon(\varepsilon)$. For this reason, we must characterize the aggregate endogenous state not only by moments of the capital distribution but also by the distribution of firms across productivity transitions, h, $N_\varepsilon \times N_\varepsilon$ arrays, where each element $h(\varepsilon_i, \varepsilon_j)$ represents the mass of firms that transitioned from productivity state ε_j in the previous period (-1) to productivity state ε_i in the current period.[13] Importantly, distribution of firms across productivity transitions embodies the information of the current and previous period marginal distribution of firm over productivity, as well as the information of the conditional transitional probability for idiosyncratic productivity, $\Pi_h^\varepsilon(\varepsilon_i|\varepsilon_j)$.[14] Accordingly, the main problem to solving granular heterogeneous firm model is the construction a feasible set of distribution of firms across productivity transitions, H, and its transition probability, Π^h. To make the problem computationally tractable, we approximate this process with a finite-state Markov chain.[15] In particular, to be consistent with the definition of granularity, we construct the set of possible aggregate productivity distributions, H, and their transition matrix, Π^h, to be consistent with the underlying ergodic productivity process, Π^ε. The key idea is to first approximate the time series of the marginal productivity distribution with a Markov chain by employing the Terry and Knotek II (2011)'s methodology, then use this to construct the transition probabilities for the joint distribution, h. The detailed description of the implementation of our numerical method to build the set of transition probability and

[13] In Appendix B, we show that we can exactly recover μ with (h, K) in a frictionless economy.

[14] We can derive the current marginal distribution of productivity, denoted by $\mu_\varepsilon(\varepsilon)$, by summing across the previous period's states: $\mu_\varepsilon(\varepsilon_i) = \sum_j h(\varepsilon_i, \varepsilon_j)$. Similarly, the previous period's marginal distribution is $\mu_{\varepsilon,-1}(\varepsilon_j) = \sum_i h(\varepsilon_i, \varepsilon_j)$. Furthermore, from the distribution of firms across productivity transitions h, we can also derive the conditional transition probability:

$$\Pi_h^\varepsilon(\varepsilon_i|\varepsilon_j) = \frac{h(\varepsilon_i, \varepsilon_j)}{\sum_f h(\varepsilon_f, \varepsilon_j)}, \quad \forall j, i = 1, 2, \ldots, N_\varepsilon. \tag{10}$$

[15] Let $H = \{h_1, \ldots, h_{N_h}\}$ be a set of N_h possible realizations of the h array. The transitions between these realizations are governed by a transition matrix Π^h, where $\Pi^h(h_l, h_m) = \Pr(h' = h_m \mid h = h_l) \geq 0$ and $\sum_{m=1}^{N_h} \Pi^h(h_l, h_m) = 1$, $\forall l = 1, 2, \ldots, N_h$.

its probability space is provided in Appendix C.

3 Quantitative Analysis

3.1 Parameters

This section describes our calibration approach. We first specify the functional forms for preferences and technology, then divide parameters into two groups: predetermined parameters set based on standard values in the literature, and calibrated parameters chosen to match key moments from both micro and macro data. We conclude by discussing how well the calibrated model fits both targeted and untargeted moments.

3.1.1 Model Specification

We assume that the representative household's period utility follows the indivisible labor specification of Hansen (1985) and Rogerson (1988): $u(c, n^h) = \ln(c) + \eta(1 - n^h)$. The firm-level production function takes a Cobb-Douglas form with decreasing returns to scale: $\varepsilon F(k, n) = \varepsilon k^\alpha n^\nu$, where $\alpha + \nu < 1$. In assuming $\alpha + \nu < 1$ we follow the span-of-control approach of Lucas (1978), an approach used in business cycle models with heterogeneous firms (e.g., Khan and Thomas, 2008; Bloom et al., 2018; Carvalho and Grassi, 2019). Under perfect competition, this curvature keeps optimal firm size finite — as $\alpha + \nu$ approaches one, the most productive firms would grow too large relative to the data — and we calibrate α jointly with the parameters of the productivity process to replicate the US firm size distribution.[16]

Productivity is the product of two components: $\varepsilon = \xi \cdot \epsilon$, where ξ is a persistent base component and ϵ is a transitory component. In each period, the base component ξ is re-

[16] Alternatively, decreasing returns in revenue can arise from constant returns in production combined with an isoelastic demand curve. Absent mechanisms that endogenize productivity or demand, the two formulations yield the same revenue function.

drawn with probability χ from a Pareto distribution $\mathcal{P}(e_m, e)$.[17] With probability $(1-\chi)$, the base component remains unchanged. The transitory part follows a log-normal process: $\ln(\epsilon) \sim \mathcal{N}(0, \sigma_\epsilon^2)$. We discretize the base component into N_ξ grid points, $\xi \in \{\xi_1, \ldots, \xi_{N_\xi}\}$, with probabilities implied by the Pareto distribution, and the transitory component into N_ϵ grid points, $\epsilon \in \{\epsilon_1, \ldots, \epsilon_{N_\epsilon}\}$, with probabilities implied by the lognormal distribution, and construct $\varepsilon = \xi \cdot \epsilon$ on the resulting $N_\varepsilon = N_\xi \times N_\epsilon$ finite grid points, $\varepsilon \in \{\varepsilon_1, \ldots, \varepsilon_{N_\varepsilon}\}$, together with the associated transition matrix $Pr(\varepsilon' = \varepsilon_n | \varepsilon = \varepsilon_m) \equiv \Pi^\varepsilon(\varepsilon_m, \varepsilon_n)$ — the Markov chain set up in Section 2.1. Conditional on the current pair (ξ_j, ϵ_h), and thus on current productivity ε_m, the distribution of next period's pair (ξ_k, ϵ_l), and thus of ε_n, is fully determined: the base component either remains at its current value, with probability $1-\chi$, or is redrawn independently of history, with probability χ, while the transitory component is drawn i.i.d.[18]

In the computation, we use the primitive chain Π^ε — which does not depend on the aggregate state — to simulate the M firms and construct the finite set of aggregate states μ together with its transition distribution G (Section 2.4 and Appendix C provide the details). Because M is finite, realized transition fractions differ from their ergodic counterparts, and conditioning on a realized aggregate transition (μ, μ') yields the state-dependent transition probabilities $\Pi^\varepsilon_{\mu'|\mu}$ of Section 2.1. The two objects are consistent: averaging across realizations of the aggregate state recovers the time-invariant chain, $\sum_{\mu'} \Pi^\mu(\mu'|\mu)\, \Pi^\varepsilon_{\mu'|\mu}(\varepsilon_i, \varepsilon_j) = \Pi^\varepsilon(\varepsilon_i, \varepsilon_j)$.

The assumption that ξ is redrawn from a Pareto distribution with probability χ serves a dual purpose in our model. First, it generates the fat-tailed firm size distribution observed empirically, similar to Ehouarne, Kuehn and Schreindorfer (2022).[19] Second, it captures the

[17] The Pareto distribution $\mathcal{P}(e_m, e)$ has probability density function $f(\xi) = \frac{e \cdot e_m^e}{\xi^{e+1}}$ for $\xi \geq e_m$, where $e_m > 0$ is the scale parameter (minimum value) and $e > 0$ is the shape parameter determining the tail thickness.

[18] The transition matrix is $\Pi^\varepsilon(\varepsilon_m, \varepsilon_n) = \Pi^\varepsilon\big[(\xi_j, \epsilon_h) \to (\xi_k, \epsilon_l)\big] = \big[(1-\chi)\, \mathbf{1}_{\{\xi_k = \xi_j\}} + \chi\, p^P(\xi_k)\big] \cdot p^{LN}(\epsilon_l)$, where $\varepsilon_m = \xi_j \cdot \epsilon_h$, $\varepsilon_n = \xi_k \cdot \epsilon_l$, and p^P and p^{LN} denote the Pareto and lognormal probabilities on the grids. The N_ε productivity levels are all distinct, so each ε_m identifies a unique pair (ξ_j, ϵ_h). We set $N_\xi = 7$ and $N_\epsilon = 3$, so that $N_\varepsilon = 21$.

[19] Ehouarne et al. (2022) shows that permanent idiosyncratic shocks combined with a power law distribution in firm size are crucial for generating time-variation in common idiosyncratic skewness (see also Salgado, Guvenen and Bloom (2019)).

pervasive micro-level rank reversals among firms as even major corporations can experience dramatic sales declines, "micro disasters," that fundamentally alter their market position (Autor, Dorn, Katz, Patterson and Van Reenen, 2020). This modelling approach is also consistent with recent evidence by Melcangi and Sarpietro (2024), who document that firm-level sales and productivity are hit by heavy-tailed shocks.

Table 1: Moments

Targeted Moments			
Description	Parameter Sensitivity	Data	Model
Capital-Output Ratio ($\frac{K}{Y}$)	α, e_m, χ	2.300	2.23
Total Hours Worked (N)	η	0.333	0.336
TOP 100 Std. Dev. $\Delta\varepsilon$ (perc.)	e, e_m	12.00	11.90
Sales share TOP 500 firms (perc.)	e_m	40.00	39.50
Walmart Employment Share (perc.)	e, e_m	1.000	1.087
$Pr(\lvert\frac{i}{k}\rvert \leq 0.01)$	ψ, α, e	$0.08 - 0.111$	0.097
Untargeted Moments			
Tail index of Firm size dist.		1.097	1.078
Sales share TOP 50 Firms (perc.)		24.00	22.40
Sales share TOP 100 Firms (perc.)		29.00	27.55
$Pr(\frac{\Delta k'}{k} \leq -0.2)$ previous TOP 500 firms		0.024	0.022

Note: The "Parameter Sensitivity" column indicates which parameters primarily affect each targeted moment, not a one-to-one mapping.

3.1.2 Calibration Strategy

We set the length of a period in the model to be one year. The parameters fall into two categories:

Predetermined parameters: We set five parameters based on standard values in the literature. The household discount factor, $\beta = 0.96$, implies an average real interest rate of four percent (Gomme, Ravikumar and Rupert, 2011). The labor share, $\nu = 0.60$, matches the average labor share of income (Cooley, 1995). The depreciation rate of capital, $\delta = 0.07$, targets the average investment-to-capital ratio. The standard deviation of the transitory component, $\sigma_\epsilon = 0.022$, follows (Khan and Thomas, 2008). Finally, we set the number of

firms $M = 4.5$ million to match the approximate number of firms in the US economy.

Table 2: Parameter values

Pre-determined parameters		
Description	Parameter	Value
Discount factor	β	0.96
Labor share	ν	0.60
Depreciation rate	δ	0.07
Std. dev. transitory component	σ_ϵ	0.022
Number of firms	M	4.5 mil.
Calibrated parameters		
Capital share	α	0.26
Preference parameter	η	2.15
Probability of reset	χ	0.02
Pareto scale parameter	e_m	6.89
Curvature of Pareto distribution	e	6.90
Investment irreversibility	ψ	0.35

Calibrated parameters: The remaining six parameters are jointly calibrated to match key moments from the data. These parameters are: (1) the capital share, α, (2) the preference parameter, η, (3) the shape parameter of the Pareto distribution, e, (4) the scale parameter of the Pareto distribution, e_m, (5) the probability of resetting the base component, χ, and (6) the degree of investment irreversibility ψ.

We target the following six data moments: (1) the aggregate capital-output ratio of 2.3 as the average private capital-to-output ratio between 1954 and 2002 reported by Khan and Thomas (2013), (2) aggregate total hours worked of one-third, (3) standard deviation of firm-level TFP growth among the 100 largest firms of 12% (Gabaix, 2011), (4) the sales share of largest 500 firms of 40% (Gabaix, 2011), (5) the largest firm's employment share of 1% (Walmart's domestic employment size of 1.4 million as in (Carvalho and Grassi, 2019)), and (6) inaction rate defined as the mass of firms among the 100 largest firms of the previous period whose $|i/k| \leq 0.01$.[20]

[20]We target an intermediate value between 0.080 of Khan and Thomas (2008) and 0.111 obtained from Compustat. The investment rate is given by the ratio of capital expenditures (CAPX) to lagged PPEGT after controlling for SECTOR $\times$ YEAR. See Appendix D for further details on variable construction.

3.1.3 Results

Tables 1 and 2 present our calibration results. Table 1 shows both targeted and untargeted moments, while Table 2 reports the calibrated parameter values.

The calibration yields several important findings. First, the calibrated degree of capital irreversibility ($\psi = 0.35$) aligns closely with the empirical findings of Bloom (2009). Second, the reset probability $\chi = 0.02$ generates realistic firm dynamics. As Autor et al. (2020) document, the probability that a firm remains in the top 500 after one year is 0.95, which our model matches precisely with $1 - \chi = 0.98$ for annual transitions. Third, the model successfully replicates the firm size distribution. The estimated tail index (1.078) closely matches the empirical value of 1.097 reported by Carvalho and Grassi (2019). Furthermore, the model accurately reproduces the right tail of the distribution: the output shares of the top 50 and top 100 firms are very close to the empirical estimates provided by Gabaix (2011). The model's ability to replicate the skewness of the firm size distribution is particularly important, as this feature is essential for generating granular volatility—an insight originally emphasized by Gabaix (2011). The strong performance on untargeted moments provides validation of our modelling approach. These moments—particularly the tail index and sales concentration measures—were not directly targeted in our calibration but emerge naturally from the model's structure. This suggests that our productivity process with Pareto-distributed resets captures fundamental features of firm dynamics. Finally, the model replicates the negative net investment spikes observed among the largest firms in the economy, indicating that the calibrated stochastic process generates realistic micro-level investment dynamics.

The model closely matches the targeted capital–output ratio of 2.3, and the calibrated capital share is approximately what the user cost of capital would imply without the friction: with a real interest rate of four percent and a depreciation rate of seven percent, $(r+\delta)K/Y \approx 0.26$. Given the labor share of $\nu = 0.60$, the residual $1 - \alpha - \nu = 0.14$ of output constitutes pure profits, accruing to the household as the owner of firms. A profit share of this size

would have been at odds with the small economic profits measured on 1959–89 data by Basu and Fernald (1997); it is, however, in line with the substantial rise in the measured profit share documented since then (Barkai, 2020; De Loecker, Eeckhout and Unger, 2020).[21]

3.2 Business Cycles

We now examine how capital irreversibility shapes key business cycle dynamics in our granular economy, distinguishing between its direct impacts on firm behavior and its broader general equilibrium consequences.

Before examining the specific channels through which capital irreversibility operates, it is worth noting that our granular economy generates business cycle dynamics similar to those of a typical real business cycle model with aggregate TFP shocks. As shown in Table 3, the model captures key empirical regularities: investment is substantially more volatile than output ($\sigma(\ln I)/\sigma(\ln Y) = 4.102$), consumption is less volatile than output (0.578 versus 0.800), and there are strong positive contemporaneous correlations with output in consumption, investment, and total hours worked. These patterns emerge from granular shocks—idiosyncratic productivity fluctuations at large firms—rather than aggregate TFP shocks, consistent with the seminal findings of Carvalho and Grassi (2019), who show that granular shocks can generate realistic business cycle dynamics.

In levels, however, granular shocks alone fall short of the observed volatility of output: the model generates about a quarter of its standard deviation in the data (0.294 versus 1.299 percent). The natural extension would be to add an aggregate TFP shock and let the two sources of fluctuations jointly match the observed volatility. Solving the model with both sources of aggregate uncertainty is, however, beyond what is computationally feasible for us: the aggregate state tracks the realized distribution of firms across productivity transitions, and an aggregate shock would multiply this state space by its own grid. We leave this

[21] Basu and Fernald (1997) impute a required return to capital in 1959–89 industry data and estimate pure profits of at most three percent of revenue; Barkai (2020) applies the same accounting as a time series and finds a rise of 13.5 percentage points in the profit share between 1984 and 2014, while De Loecker et al. (2020) estimate that the average profit rate rose from about one to about eight percent of sales over 1980–2016.

extension for future work.

Another thing to note is that the model generates real interest rate dynamics different from a typical real business cycle model with aggregate TFP shocks. Also, the correlation of the real interest rates with output depends on the degree of capital irreversibility.[22] Without irreversibility ($\psi = 0.00$), the real interest rate is essentially acyclical ($\rho(\ln(1+r), \ln Y) = -0.022$), while with irreversibility ($\psi = 0.35$), it becomes moderately counter-cyclical ($\rho(\ln(1+r), \ln Y) = -0.493$). This sensitivity to capital market frictions reflects how irreversibility constrains both firms' capital adjustments and households' consumption smoothing, affecting equilibrium interest rate determination through precautionary savings motives.[23]

[22] Given the aggregate state μ, the real interest rate is computed as the expected consumption growth: $r = \frac{\sum_{\mu'} \Pi^{\mu}(\mu'|\mu) C'(\mu')}{C(\mu)} - 1$. As shown in Appendix C.2, by approximating the endogenous aggregate state with moments of the endogenous marginal distribution of k and h, we derive C' using a log-linear forecasting rule.

[23] Appendix E, Table E.1, shows that the real interest rate is countercyclical when considering the 1964–2019 sample. Although it appears procyclical in the 1980–2019 sample, the degree of procyclicality remains substantially lower than that predicted by standard real business cycle models, wherein persistent improvements in technology typically induce simultaneous increases in consumption and investment demand, which, in the short run, outpace supply and generate procyclical movements in the real interest rate (Beaudry and Guay, 1996). See also Winberry (2021) for the recent contribution on the cyclicality of the real interest rates in a real business cycle model.

Table 3: Business Cycle Moments and Capital Irreversibility

Variable	Statistic	$\psi = 0.350$	$\psi = 0.234$	$\psi = 0.117$	$\psi = 0.000$	DATA
	$\sigma(x)$	0.294	0.327	0.362	0.377	1.299
$\ln Y$	$\sigma(x)/\sigma(Y)$	1.000	1.000	1.000	1.000	1.000
	$\rho(x, \ln Y)$	1.000	1.000	1.000	1.000	1.000
	$\sigma(x)$	0.170	0.156	0.140	0.119	1.039
$\ln C$	$\sigma(x)/\sigma(Y)$	0.578	0.477	0.387	0.316	0.800
	$\rho(x, \ln Y)$	0.785	0.771	0.747	0.716	0.899
	$\sigma(x)$	1.208	1.452	1.751	1.962	5.913
$\ln I$	$\sigma(x)/\sigma(Y)$	4.102	4.448	4.837	5.209	4.552
	$\rho(x, \ln Y)$	0.903	0.940	0.962	0.975	0.916
	$\sigma(x)$	0.192	0.229	0.274	0.303	1.700
$\ln H$	$\sigma(x)/\sigma(Y)$	0.652	0.701	0.756	0.805	1.309
	$\rho(x, \ln Y)$	0.836	0.901	0.941	0.962	0.881
	$\sigma(x)$	0.025	0.022	0.020	0.018	1.310
$\ln(1 + r)$	$\sigma(x)/\sigma(Y)$	0.085	0.068	0.056	0.049	1.007
	$\rho(x, \ln Y)$	-0.493	-0.390	-0.237	-0.022	-0.286

Notes: The table shows model and data business cycle moments of output Y, consumption C, investment I, hours worked H, and the real interest rate r, under alternative degrees of capital irreversibility indexed by ψ. All model columns correspond to the same general-equilibrium granular economy; only the irreversibility parameter varies across specifications, with intermediate values $\psi \in \{0.234, 0.117\}$ chosen as equal thirds between the benchmark ($\psi = 0.350$) and the frictionless ($\psi = 0.000$) economies. $\sigma(x)$ is the standard deviation of x, $\sigma(x)/\sigma(Y)$ the standard deviation relative to that of Y, and $\rho(x, \ln Y)$ the contemporaneous correlation of x with $\ln Y$. Model moments are obtained from a 15,000-period unconditional simulation using the solution of the model. All series are HP-filtered in logs with a smoothing parameter of 6.25, following Ravn and Uhlig (2002). Data sources for the reported moments are: (1) real gross domestic product (GDPCA taken from FRED), (2) real personal consumption expenditures (PCECCA taken from FRED), (3) real gross private domestic investment (GPDICA taken from FRED), (4) total nonfarm business sector hours (HOANBS taken from FRED but annualized), and (5) nominal return on 1-year Treasury bills (DGS1 taken from FRED) adjusted for realized CPI inflation (CPIAUCSL_PC1 taken from FRED). The sample refers to the period spanning from 1964 to 2019.

3.2.1 Direct Effects on Investment and Capital Allocation

To understand this result, we first decompose the mechanisms at play, beginning with the direct effects of irreversibility on firm-level behavior. Capital irreversibility (ψ) plays a significant role in shaping aggregate volatility in our granular economy, primarily through its *direct effects* on firm-level investment and capital allocation. As evidenced in Table

3, overall output volatility, $\sigma(\ln Y)$, decreases as the degree of capital irreversibility (ψ) increases, falling from 0.377 in the frictionless model ($\psi = 0.00$) to 0.294 in the benchmark model with irreversibility ($\psi = 0.35$).[24]

This can be understood through key *direct mechanisms*. Higher irreversibility causes firm-level capital stocks to adjust more sluggishly to idiosyncratic productivity shocks. This reduced responsiveness is a key channel, leading to a lower covariance between capital and current productivity.[25] In a granular economy, such a dampened alignment of capital with productivity contributes to reducing aggregate output volatility. This dampened alignment is evident in our simulation results: the average log dispersion of the marginal product of capital ($\ln \sigma^{MPK}$) increases from 0.019 in the frictionless model to 0.022 in our benchmark model with irreversibility. Investment fluctuations are also dampened under irreversibility. Table 3 shows that aggregate investment volatility, $\sigma(\ln I)$, drops from 1.962 when $\psi = 0.00$ to 1.210 when $\psi = 0.35$, rendering investment substantially less responsive to shocks.

3.2.2 The Role of Granularity: A Within-Model Comparison

The propagation mechanism documented above arises specifically from the interaction between irreversibility and granular shocks. To establish this within our own framework, we compare the benchmark economy with one in which aggregate fluctuations are driven exclusively by aggregate TFP shocks. Table 4 reports business cycle moments for an economy calibrated to the same ergodic distribution as our benchmark, but in which volatility arises solely from an aggregate TFP process calibrated as in Khan and Thomas (2008). In this environment, variations in ψ have only negligible effects on aggregate investment and output

[24]Eliminating partial investment irreversibility can have two countervailing effects on the shape coefficient of the firm size distribution. On the one hand, irreversibility limits firms' ability to shrink, creating persistence and path dependence in firm size; removing this friction allows less productive firms to contract more readily, thinning the upper tail and increasing the shape coefficient. On the other hand, greater responsiveness to shocks in the frictionless economy may increase firm-level volatility, which tends to fatten the tail and reduce the shape coefficient. In our model, the first effect dominates: the shape coefficient in the frictionless economy is 1.139, indicating a thinner-tailed, more equal firm size distribution.

[25]Recall Appendix A: aggregate output volatility is increasing in the productivity–capital covariance $\sigma_{\varepsilon k}$, which is measurable through the cross-sectional dispersion of marginal products of capital.

volatility: the irreversibility constraint is not cyclically engaged when shocks are common to all firms. This echoes a familiar finding in general equilibrium models with aggregate shocks (Veracierto, 2002; Thomas, 2002; Khan and Thomas, 2008).

Table 4: Business Cycle Moments: Aggregate TFP Shock

	$\psi = 0.35$ with $y = Z\varepsilon k^{\alpha} l^{\nu}$			$\psi = 0.00$ with $y = Z\varepsilon k^{\alpha} l^{\nu}$		
	$\sigma(x)$	$\frac{\sigma(x)}{\sigma(Y)}$	$\rho(x, \ln Y)$	$\sigma(x)$	$\frac{\sigma(x)}{\sigma(Y)}$	$\rho(x, \ln Y)$
$\ln Y$	1.888	1.000	1.000	1.898	1.000	1.000
$\ln C$	0.981	0.520	0.924	0.957	0.504	0.934
$\ln I$	6.775	3.589	0.963	7.412	3.905	0.965
$\ln H$	1.051	0.557	0.934	1.062	0.559	0.946
$\ln(1+r)$	0.169	0.089	0.848	0.151	0.079	0.871

Notes: The table compares the business cycle moments of model economies whose volatility is exclusively driven by aggregate TFP shock. In particular, $\psi > 0$ refers to the economy with partial capital irreversibility, and $\psi = 0$ refers to the economy without partial irreversibility. $\sigma(x)$ is the percentage standard deviation of x, and $\sigma(x)/\sigma(\ln Y)$ is the relative standard deviation to that of Y, and $\rho(x, \ln Y)$ is the contemporaneous correlation of x with Y. All series are HP-filtered with a smoothing parameter of 6.25, following Ravn and Uhlig (2002).

This aggregate-shock economy can be interpreted as the limiting case of our framework as the number of firms grows large. As the cross-section expands, idiosyncratic shocks average out by the law of large numbers, granular fluctuations vanish, and the economy converges to one in which aggregate uncertainty is driven solely by common shocks. The volatility reduction documented in Table 3 is therefore attributable to the interaction between irreversibility and granular shocks, not to irreversibility per se.

3.2.3 Indirect General Equilibrium Effects

Beyond its direct effects on firm-level investment, irreversibility shapes aggregate dynamics through an indirect general equilibrium channel. When capital adjustment is constrained,

the margin firms would normally use to cushion shocks is unavailable, and adjustment is forced onto factor and asset markets in general equilibrium. This manifests on two margins.

The first is the consumption-savings margin. Aggregate consumption is more volatile under irreversibility—$\sigma(\ln C)$ rises from 0.119 in the frictionless model to 0.170 in the benchmark (Table 3)—and the real interest rate moves from essentially acyclical in the frictionless model ($\rho(\ln(1+r), \ln Y) = -0.022$) to countercyclical in the benchmark (-0.493). These patterns are familiar from a literature on asset prices in production economies, where they have been generated through habit formation on the household side (Campbell and Cochrane, 1999) or through capital adjustment and investment frictions on the production side (Jermann, 1998; Beaudry and Guay, 1996; Winberry, 2021; Chen, 2017). Our results sit naturally within this body of work.

The second is the labor margin, which is central to the output-volatility result. The wage response follows from the household's intratemporal optimality condition,

$$\omega(\mu) = \eta C(\mu), \tag{11}$$

which links the equilibrium real wage one-for-one to aggregate consumption. With $\sigma(\ln C)$ elevated, real wages are correspondingly more volatile and fall more sharply in downturns. This acts as a shock absorber for firms: a larger wage reduction moderates the fall in labor demand, attenuating the decline in employment and production. The volatility of hours worked drops from 0.303 in the frictionless model to 0.193 in the benchmark.

The reduction in aggregate output volatility thus reflects two channels: the direct effect on firm-level capital allocation (Section 3.2.1) and the indirect wage channel just described. Quantitatively, the indirect channel dominates: it accounts for roughly three-quarters of the total reduction in $\sigma(\ln Y)$, with the direct channel responsible for the remainder.[26] The elevated consumption volatility and countercyclical real rate documented above are reflections

[26]The decomposition isolates the direct channel via a counterfactual in which capital adjustment costs shape firms' decisions but entail no real resource reallocation, muting the general equilibrium feedback through consumption. Under this counterfactual, $\sigma(\ln Y)$ falls by only 5% relative to the frictionless benchmark, against a 22% reduction in the full benchmark. See Appendix F (Table F.1) for details.

of the same underlying constraint, not independent outcomes.

3.3 Level Effects of Irreversibility

Irreversibility not only shapes how the economy fluctuates cyclically but also shapes the level of aggregate activity. For instance, irreversibility decreases the optimal level of capital stock of firms as they anticipate that the constraint may bind in the future, which leads to costly disinvestment (ex-ante effect). On the other hand, even when productivity falls firms may end up holding more capital stock than they would choose, because it is costly to disinvest under the irreversibility constraint (ex-post effect). In theory, therefore, it is ambiguous whether the ex-ante effect dominates the ex-post effect, or the other way around, and this subsection studies the level of aggregate variables with or without investment irreversibility and the same step-by-step simulation procedure used above.[27]

Table 5 reports time averages from the simulations: all economies are run along a common sequence of idiosyncratic shocks, so the columns differ in firms' decision rules and in the law of motion — that is, in how much of general equilibrium is allowed to operate. Column *(i)* evaluates frictionless decision rules at benchmark prices, and there the capital stock is 15.3 percent higher than in the Benchmark. Removing the friction raises gross disinvestment by 64 percent and gross investment by 24 percent; but since gross investment is five times larger in level, the extra investment exceeds the extra disinvestment in units of capital, and the capital stock rises. Once the goods market clears period by period as in Column *(ii)*, the increase in capital falls from 15.3 to 6.4 percent. Allowing expectations to adjust as well changes little, 6.0 percent as seen in Column *(iii)*. The capital stock is higher without the friction both at fixed prices and in full equilibrium: the ex-ante effect dominates the ex-post effect.[28]

Output is higher without the friction for two reasons: the economy operates with more

[27] We are grateful to the referee for suggesting this exercise in this granular economy.

[28] See Abel and Eberly (1999) for the theoretical ambiguity of how irreversibility impacts the level of capital stock.

capital, and capital is better allocated across firms — the dispersion of marginal products of capital falls from 0.022 to 0.019, already at fixed prices. In full equilibrium, output is 1.3 percent above the Benchmark. The labor response is more involved. At benchmark prices, firms with frictionless decision rules demand 10.1 percent more labor. In equilibrium, however, the wage moves one-for-one with consumption through the intratemporal condition $\omega(\mu) = \eta C(\mu)$ of Section 3.2.3. Consumption is higher without the friction, the wage rises accordingly, and hours end up marginally below their Benchmark level. The additional output is thus produced with no additional hours.

Table 5: Level Effects of Irreversibility: Aggregates

		Frictionless firms ($\psi = 0$)		
	Benchmark $\psi = 0.35$	(i) Fixed prices	(ii) Market clearing	(iii) Full equilibrium
Capital K	100 (1.730)	115.26	106.36	105.95
Gross investment $\int \hat{i}^+ \, dm$	100 (0.151)	123.91	114.30	113.91
Gross disinvestment $\int \hat{i}^- \, dm$	100 (0.029)	163.79	151.03	151.72
Output Y	100 (0.777)	110.17	101.42	101.29
Labor N	100 (0.336)	110.12	99.40	99.11
$\ln \sigma^{\text{MPK}}$	0.022	0.019	0.019	0.019

Notes: Aggregate quantities are normalized to 100 in the Benchmark economy, with their levels in parentheses. The Benchmark and column *(iii)* are the equilibria underlying the $\psi = 0.350$ and $\psi = 0.000$ columns of Table 3, respectively. In columns *(i)* and *(ii)*, firms solve their problem with $\psi = 0$ under the benchmark's converged forecasting rules. In *(i)*, these decision rules are obtained at the price that equilibrates the benchmark simulation each period (so the same price sequence as in the benchmark case). In *(ii)*, the goods market clears period by period, while firms form expectations with the same benchmark forecasting rules. The residual step to *(iii)* is the adjustment of expectations, so with the converged forecasting rule as in Table 3. All economies are simulated under the same sequence of idiosyncratic shocks.

3.4 Dynamics of Micro-level Investment Moments

The preceding analysis established that capital irreversibility dampens aggregate output volatility in our granular economy, and that this effect arises specifically from the interaction between irreversibility and granular shocks (Section 3.2.2). To understand *why* this interaction matters, we now turn to the micro-level investment behavior that underlies the aggregate result. Table 6 presents key statistics on firms' investment dynamics, compar-

ing our benchmark granular economy with a counterpart economy in which fluctuations are driven solely by aggregate TFP shocks.

A critical distinction emerges from Table 6 when observing the cyclical behavior of the irreversibility constraint itself, especially for large firms. In the granular economy, the share of TOP 500 firms constrained by irreversibility ($Pr((1-\delta)k \geq k_u^\star)$) —meaning they wish to downsize their capital stock but cannot due to the constraint—exhibits notable volatility ($\sigma(x) = 0.8$ percentage points) and is strongly counter-cyclical ($\rho(x, \ln Y) = -0.357$). During downturns, which can be initiated or amplified by adverse granular shocks hitting these large firms, a significant and cyclically varying number of economically important firms are actively impeded by irreversibility.

By contrast, in the aggregate-shock economy, the share of TOP 500 firms constrained by irreversibility has virtually no volatility ($\sigma(x) = 0.0$) and a much weaker correlation with the aggregate cycle. While some large firms are constrained on average, this fraction does not fluctuate meaningfully over the business cycle in response to aggregate shocks alone. This lack of cyclical variation is why, in this aggregate-shock counterpart, irreversibility has no active margin through which to shape the cycle.

Table 6: Micro Investment Behavior (Values for x and $\sigma(x)$ in percent)

	Granular Economy			Aggregate-Shock Economy		
Micro-moment	x	$\sigma(x)$	$\rho(x, \ln Y)$	x	$\sigma(x)$	$\rho(x, \ln Y)$
$Pr(\lvert\frac{i}{k}\rvert \leq 0.01)$	9.7	0.2	0.060	8.9	0.2	0.318
$Pr(\lvert\frac{i}{k}\rvert \leq 0.01)$ TOP 500	5.4	1.2	-0.260	5.4	0.2	-0.018
$Pr(\frac{\Delta k'}{k} \leq -0.20)$	0.4	0.0	0.184	0.4	0.0	-0.295
$Pr(\frac{\Delta k'}{k} \leq -0.20)$ TOP 500	2.2	0.4	-0.525	2.2	0.0	-0.518
$Pr(i \leq 0.000)$	0.7	0.0	0.049	1.6	0.0	-0.696
$Pr(i \leq 0.000)$ TOP 500	3.5	1.7	-0.123	3.3	0.6	0.021
$Pr((1-\delta)k \geq k_u^\star)$	10.3	0.0	0.106	10.3	0.0	-0.183
$Pr((1-\delta)k \geq k_u^\star)$ TOP 500	7.7	0.8	-0.357	7.7	0.0	-0.063

Notes: The table compares the ergodic value and cyclicality of micro investment behavior in the benchmark model (Granular Economy) with a counterpart economy in which volatility is driven solely by aggregate TFP shocks. For each specified micro-moment (investment behavior): The column headed 'x' reports the mean probability, expressed in percent. The column headed '$\sigma(x)$' reports the standard deviation of this probability, expressed in percentage points. The column headed '$\rho(x, \ln Y)$' reports the contemporaneous correlation of the micro-moment with the logarithm of aggregate output Y. Investment behaviors considered are: inaction rate ($Pr(\lvert\frac{i}{k}\rvert \leq 0.01)$), negative net investment spike rate ($Pr(\frac{\Delta k'}{k} \leq -0.20)$), negative investment rate ($Pr(i \leq 0.000)$), and the probability of the irreversibility constraint binding ($Pr((1-\delta)k \geq k_u^\star)$). These are calculated for both the entire economy and for the largest 500 firms from the previous period.

This difference in the cyclical engagement of the irreversibility constraint among large firms is fundamental. When granular shocks hit, capital irreversibility actively shapes the investment response of a cyclically varying segment of the economy's largest firms. Their inability to disinvest optimally during periods of idiosyncratic distress means capital remains temporarily misallocated, but it also means aggregate investment does not contract as sharply as it otherwise would, and productive resources are not shed as precipitously. The time-varying incidence of binding irreversibility constraints among large firms is the micro-level mechanism through which granular shocks and irreversibility interact to shape aggregate dynamics.[29]

[29]Further evidence from Table 6 supports this. For instance, the inaction rate ($Pr(\lvert\frac{i}{k}\rvert \leq 0.01)$) among the TOP 500 firms is more volatile ($\sigma(x) = 1.2$ percentage points vs. 0.2 percentage points) and more strongly counter-cyclical ($\rho(x, \ln Y) = -0.260$ vs. -0.018) in the granular economy. Similarly, large disinvestments ($Pr(\frac{\Delta k'}{k} \leq -0.20)$) by TOP 500 firms are also more volatile and counter-cyclical in the presence of granular shocks.

4 Welfare and Policy Analysis

We now evaluate the welfare consequences of capital irreversibility and of a widely advocated policy response: a government subsidy that reimburses firms for their liquidation losses. Such schemes — scrappage programs, accelerated depreciation allowances on disposed assets, and other forms of liquidation relief — are used in practice and directly target the loss irreversibility places on firms' decisions. The subsidy removes the friction from firms' decisions but not from the aggregate resource constraint, and firms — now compensated at the margin while the economy continues to bear the resource cost — disinvest more aggressively than before. We quantify the welfare loss from the subsidy and set it against the gain from removing the friction itself.

To see how, we construct four economies that turn on one mechanism at a time, summarized in Table 7. Each economy isolates one channel. *(1: Benchmark)* is the economy with partial irreversibility, $\psi = 0.35$. *(2-a: No resource loss)* shuts down the aggregate resource loss while firms still face the friction in their decisions, so the step *(1)* $\rightarrow$ *(2-a)* isolates the resource-loss channel. *(2-b: Frictionless)* additionally removes the friction from firms' decisions, so firms act as if capital were perfectly reversible: the step *(2-a)* $\rightarrow$ *(2-b)* isolates the misallocation channel. Together, these two steps account for the full welfare cost of irreversibility.[30] *(3: Subsidy)* restores the resource loss but lets firms decide as if $\psi = 0$, as in *(2-b)* — the government reimburses firms for their liquidation losses, financed by a lump-sum tax on the representative household. The step *(2-b)* $\rightarrow$ *(3)* diagnoses the subsidy against this decomposition.

Let $\hat{i}^+(k, s, z, m) \geq 0$ denote gross investment by investing firms and $\hat{i}^-(k, s, z, m) \geq 0$ denote gross disinvestment by disinvesting firms, so that gross investment is $\hat{i} = \hat{i}^+ - \hat{i}^-$ and investment net of the liquidation loss is $i = \hat{i}^+ - (1 - \psi)\hat{i}^-$, as defined in Section 2. The

[30] Both channels also affect the volatility of aggregate consumption, though, as discussed below, this margin contributes little to welfare at business-cycle frequencies (Lucas, 1987).

Table 7: Structure of the Four Economies

Economy	Firm decision rule	Resource loss Ψ
(1: Benchmark)	$(\hat{i}^+, \hat{i}^-)$ under friction ψ	$\psi \int \hat{i}^- \, dm$
(2-a: No resource loss)	$(\hat{i}^+, \hat{i}^-)$ under friction ψ	0
(2-b: Frictionless)	$(\hat{i}^+, \hat{i}^-)$ as if $\psi = 0$	0
(3: Subsidy)	$(\hat{i}^+, \hat{i}^-)$ as if $\psi = 0$	$\psi \int \hat{i}^- \, dm$, financed by lump-sum tax

Notes: $\hat{i}^+(k, s, z, m) \geq 0$ denotes gross investment by investing firms and $\hat{i}^-(k, s, z, m) \geq 0$ denotes gross disinvestment by disinvesting firms. The friction ψ is the fraction of capital value lost when a firm liquidates. $\Psi = \psi \int \hat{i}^- \, dm$ is the aggregate resource loss from disinvestment, zero in *(2-a)*, where the resource cost is absent by construction, and in *(2-b)*, where capital is reversible.

aggregate resource constraint is

$$C = Y - \int \hat{i}^+ \, dm + (1 - \psi) \int \hat{i}^- \, dm \;\equiv\; Y - \hat{I} - \Psi, \tag{12}$$

where $Y = z \int s k^\alpha n^\nu \, dm$ is aggregate output, $\hat{I} \equiv \int \hat{i}^+ \, dm - \int \hat{i}^- \, dm$ is aggregate gross investment, $I \equiv \hat{I} + \Psi$ is aggregate investment inclusive of the resource loss, and $\Psi \equiv \psi \int \hat{i}^- \, dm \geq 0$ is the aggregate resource loss from disinvestment. The four economies differ only in how firms choose $(\hat{i}^+, \hat{i}^-)$ and in the size of Ψ, summarized in Table 7. Table 8 reports aggregates, consumption volatility, efficiency measures, and welfare for the four economies, with welfare measured in consumption-equivalent variation (CEV) relative to *(1)*. The CEV in the first welfare row compares ergodic distributions, each economy evaluated at its own stationary distribution; the second row computes welfare along the transition from *(1)*. The first row compares the long-run welfare of the household across the four economies; the second is the gain, or loss, from moving the Benchmark economy to each of them, net of the cost of getting there. The two rows together give the long-run and short-run effects of the same policy, in an economy that has had it from the start (first row) and in one that adopts it now (second row). Appendix G describes the computation of the second row.

The resource-loss channel: (1) → (2-a). Removing the aggregate resource loss while firms still face the friction leaves MPK dispersion at 0.022, as in *(1)*, so the gain comes from the resources that would have been consumed by disinvestment now being preserved. In the

long run, consumption is 0.31% higher, and its volatility falls from $\sigma(\ln C) = 0.170$ to 0.125, reflecting that cyclical fluctuations in Ψ are no longer absorbed by the household's budget. Along the transition the household gains from the first period, as there is no capital to build. Capital in *(2-a)* is in fact 1.3% lower in the long run, so the household consumes part of the Benchmark's capital stock on the way, and the CEV, 1.61%, is above the 1.45% of the ergodic comparison. Irreversibility's resource waste is a first-order welfare cost.

The allocation channel: (2-a) → (2-b). Additionally removing the friction from firms' decisions leads firms to disinvest more (the ex-post effect), so capital is reallocated across firms and MPK dispersion falls from 0.022 to 0.019. Firms also invest more, no longer holding back for fear of costly disinvestment later (the ex-ante effect), and in the long run the economy holds 6% more capital and consumes 2% more than the Benchmark. The CEV rises from 1.61% to 1.83% along the transition, so the allocation channel adds 0.22%, against 1.04% (from 1.45 to 2.49) in the ergodic comparison. This is because economy *(2-b)* is evaluated at its own stationary distribution, where the larger capital stock is already in place and the household consumes its return; along the transition it must first forgo consumption to build it. The cost of irreversibility along the transition, 1.83%, thus decomposes into 1.61% from the resource-loss channel and 0.22% from the allocation channel.

The subsidy: (2-b) → (3). Under the policy firms decide as if $\psi = 0$, as in *(2-b)*, but the resource loss is reintroduced, now borne by the household through the lump-sum tax. Therefore, MPK dispersion remains 0.019 and TFP gains 0.154%, as in *(2-b)*. Firms disinvest about as much as in *(2-b)*, but each unit of disinvestment now destroys a fraction ψ of capital value, which firms do not internalize because they are compensated for it, and the aggregate resource loss rises to 60% above its Benchmark value. In the long run the economy holds more capital than *(2-b)* but consumes less, because part of output is lost to disinvestment. Welfare drops from 2.49% to 0.32% in the ergodic comparison: this is the resource loss, now borne in full by the household. Along the transition the CEV is −0.59%,

Table 8: Policy Counterfactual: Aggregate Results

	(1) Benchmark	(2-a) No resource loss	(2-b) Frictionless	(3) Subsidy
	Aggregates ((1) = 100)			
Consumption C	100 (0.646)	100.31	102.01	101.55
Output Y	100 (0.777)	98.84	101.29	103.22
Labor N	100 (0.336)	98.43	99.11	101.65
Gross investment $\int \hat{i}^{+}\, dm$	100 (0.151)	98.41	113.91	116.09
Gross disinvestment $\int \hat{i}^{-}\, dm$	100 (0.029)	98.98	151.72	153.49
Investment I	100 (0.131)	91.22	97.71	111.68
Resource loss Ψ	100 (0.010)	—	—	160.00
Capital K	100 (1.730)	98.73	105.95	107.98
	Consumption volatility			
$\sigma(\ln C)$	0.170	0.125	0.119	0.161
	Efficiency			
$\ln \sigma^{\mathrm{MPK}}$	0.022	0.022	0.019	0.019
ΔTFP (%)	—	0.0	0.154	0.154
	Welfare			
CEV, ergodic comparison (%)	—	1.45	2.49	0.32
CEV, transition from Benchmark (%)	—	1.61	1.83	−0.59

Notes: Aggregate quantities are normalized to 100 in economy *(1: Benchmark)*, with their levels in parentheses. The aggregate resource constraint (12) holds in all columns as $C = Y - \hat{I} - \Psi$ with $I = \hat{I} + \Psi$. $\Psi = 0$ by construction in *(2-a)* and *(2-b)*. Capital is the steady-state aggregate capital stock, related to $\hat{I}$ by the law of motion $K' = (1-\delta)K + \hat{I}$. ΔTFP is the percentage change in measured total factor productivity relative to *(1)*. CEV denotes the consumption-equivalent welfare change relative to *(1)*. The first welfare row compares ergodic distributions: each economy is evaluated at its own stationary distribution. The second computes welfare along the equilibrium transition path when the policy is imposed at date zero, unexpectedly and permanently, starting from the Benchmark's own ergodic distribution. Appendix G describes the computation.

and the further drop from 0.32% is the cost of accumulating the larger capital stock, as in the step to *(2-b)*. Consumption volatility also rises relative to *(2-b)*, from $\sigma(\ln C) = 0.119$ to 0.161, but in the tradition of Lucas (1987), this margin contributes little to welfare at business-cycle frequencies.

Taking stock. In the ergodic comparison the three steps add up to $+1.45 + 1.04 - 2.17 = +0.32\%$; along the transition, to $+1.61 + 0.22 - 2.42 = -0.59\%$. The positive ergodic number does not signal an inefficiency: the Benchmark is a competitive equilibrium without market failure, so the subsidy has nothing to correct and should reduce welfare, as the transition number shows. The case for liquidation support in practice rests on frictions absent from our model, such as financial frictions and fire-sale externalities, under which one firm's liquidation depresses the price at which others can sell. Therefore, combining granularity, irreversibility, and financial frictions in one model is a natural direction for future work.

5 Conclusion

In this paper, we presented a general equilibrium business cycle model of heterogeneous firms with investment irreversibility, in which the firm size distribution is fat-tailed and idiosyncratic shocks to large firms do not wash out in the aggregate. When aggregate fluctuations originate from firm-level shocks, irreversibility dampens the response of large firms to negative shocks, reducing the covariance between productivity and capital and, in turn, lowering output and investment volatility. These effects are amplified in general equilibrium: limited reallocation impairs households' ability to smooth consumption, increasing real wage flexibility and further absorbing shocks.

We use this model to study a popular policy response to the friction, a subsidy that compensates firms for their liquidation losses. Whereas removing the friction raises welfare, the subsidy reduces it. Policies that subsidize the symptoms of irreversibility without addressing its root causes encourage excessive capital destruction, as firms fail to internalize the social

costs of reallocation.

We believe that our results demonstrate a potential pitfall in the analysis of such policies: failure to identify a genuine market failure, and to pinpoint the root cause, may create misleading public debates on such policies. In economies where large firms drive aggregate fluctuations, policymakers should prioritize reducing frictions in capital reallocation markets — through improved bankruptcy procedures, standardization of assets, or development of secondary markets — rather than simply compensating firms for liquidation losses. Historical interventions during financial crises — such as TARP's purchase of distressed assets at above-market prices, designed to "restore liquidity and stability to the financial system" (U.S. Department of the Treasury, Office of Financial Stability, 2010) — illustrate how policies aimed at reducing capital-market frictions differ materially from subsidies that compensate for liquidation losses. Two extensions are left for future work: aggregate productivity shocks alongside granular shocks, and financial frictions, under which liquidation support may have a distortion to correct.

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